

# Gaussian Splatting and Neural Rendering for Dynamic 3D Scene Reconstruction

## Abstract

The landscape of 3D scene reconstruction and novel view synthesis has undergone a paradigm shift with the advent of 3D Gaussian Splatting (3DGS). By replacing implicit neural radiance fields with explicit, anisotropic Gaussian primitives optimized via differentiable rasterization, 3DGS has achieved real-time rendering speeds while maintaining high visual fidelity. However, extending this static representation to dynamic scenes—characterized by non-rigid deformations, complex topology changes, and large-scale motion—presents significant challenges. This survey provides a comprehensive review of the rapidly evolving field of Gaussian Splatting for dynamic 3D scene reconstruction. We systematically categorize existing methodologies into distinct families: deformation-based canonical space modeling, 4D spatiotemporal primitives, motion-factorized representations, physics-informed dynamics, and hybrid mesh-Gaussian frameworks. Furthermore, we examine specialized applications in autonomous driving, surgical navigation, and egocentric vision, alongside critical discussions on compression, streaming, and handling of adverse conditions such as motion blur and sparse views. By synthesizing insights from over 300 recent studies, we identify key trends, evaluate technical trade-offs between fidelity and efficiency, and outline open problems regarding long-term consistency, physical plausibility, and generalization in dynamic neural rendering.

## 1. Introduction

The reconstruction of dynamic 3D scenes from multi-view or monocular video sequences has long been a cornerstone problem in computer vision and graphics. Traditional approaches relied on geometric modeling or texture mapping, which often struggled with complex non-rigid motions and view-dependent effects. The introduction of Neural Radiance Fields (NeRF) marked a significant advancement by representing scenes as continuous volumetric functions parameterized by Multi-Layer Perceptrons (MLPs) [1], [2]. While NeRF achieved photorealistic novel view synthesis, its reliance on volumetric ray-marching resulted in prohibitive training and rendering times, limiting its applicability to dynamic scenarios requiring real-time interaction [3], [4].

The emergence of 3D Gaussian Splatting (3DGS) fundamentally altered this trade-off. By representing scenes as collections of explicit 3D Gaussian ellipsoids and utilizing a tile-based differentiable rasterizer, 3DGS enables real-time rendering at high resolutions on consumer hardware [5], [3], [6]. The core insight is that explicit point-based representations avoid the empty space queries inherent in volumetric methods, allowing for efficient gradient-based optimization of position, covariance, opacity, and spherical harmonics coefficients [5], [4]. However, the original 3DGS formulation assumes a static scene, making it ill-suited for dynamic environments where geometry and appearance evolve over time.

Extending 3DGS to dynamic settings requires addressing several fundamental challenges: modeling temporal coherence without exploding parameter counts, handling topological changes such as object separation or merging, ensuring physical plausibility of motion, and managing the computational load of time-varying attributes. Recent literature has proposed a diverse array of solutions, ranging from attaching deformation fields to a canonical static set of Gaussians [7], [8] to explicitly modeling 4D spatiotemporal primitives [9], [10]. Other approaches leverage motion factorization [11], physics-based constraints [12], [13], or hybrid representations combining Gaussians with meshes [14], [15] to improve stability and editability.

This survey synthesizes the state-of-the-art in dynamic Gaussian Splatting. We organize the field into coherent methodological systems, analyzing how different architectures address the spatio-temporal modeling problem. We explore the spectrum from deformation-based methods that warp a canonical space to fully 4D representations that encode time directly into primitive attributes. We further discuss specialized domains including autonomous driving [16], [17], surgical reconstruction [18], [19], and egocentric vision [20], [21], highlighting domain-specific constraints and innovations. Finally, we address critical issues such as model compression [22], [23], streaming reconstruction [24], [25], and robustness to artifacts like motion blur [26], [27]. By integrating findings from a broad range of recent works, this paper aims to provide a structured understanding of current capabilities and future directions in dynamic neural rendering.

## 2. Method

The methodological landscape of dynamic 3D Gaussian Splatting can be broadly categorized based on how temporal information is integrated into the scene representation. We identify five primary paradigms: (1) Canonical Space Deformation, where a static set of Gaussians is warped by time-conditioned fields; (2) 4D Spatiotemporal Primitives, which extend Gaussian attributes to include time; (3) Motion Factorization and Sparse Control, which decompose complex motions into shared bases or control points; (4) Physics-Informed and Mesh-Hybrid approaches, which impose physical or topological constraints; and (5) Specialized Architectures for specific domains like driving or surgery.

### 2.1 Canonical Space Deformation Fields

A dominant strategy for dynamic reconstruction involves defining a static “canonical” space containing a set of 3D Gaussians and learning a deformation field that maps these canonical points to their observed positions at any given timestamp. This approach leverages the efficiency of static 3DGS while delegating temporal complexity to a compact neural network.

**2.1.1 Global and Local Deformation Networks** The foundational work in this category, Deformable 3DGS, decouples static Gaussians in a canonical space from a time-conditioned MLP deformation field [7]. This MLP inputs the canonical position and time to predict offsets for center, rotation, and scale. To prevent overfitting and ensure smoothness, techniques such as annealing smooth training with decaying temporal noise are employed [7]. Similarly, E-D3DGS replaces coordinate-based inputs with per-Gaussian latent embeddings, allowing the deformation network to learn unique motion characteristics for each primitive independent of spatial grid resolution [28]. This decoupling enables the model to capture fine-grained details that global coordinate-based fields might smooth over.

To address the computational burden of evaluating deformation networks for millions of Gaussians, Superpoint Gaussian Splatting (SP-GS) clusters Gaussians into superpoints based on rigid motion similarity [29]. A tiny MLP then predicts deformations at the superpoint level, drastically reducing the number of network evaluations while maintaining high fidelity. This aggregation strategy is particularly effective for scenes with semi-rigid components. Further enhancing efficiency, Speedy Deformable 3DGS (SpeeDe3DGS) integrates temporal sensitivity pruning and GroupFlow, replacing per-Gaussian neural evaluations with grouped motion predictions and removing low-contribution Gaussians via Hessian-based scores [30].

**2.1.2 Hierarchical and Motion-Aware Partitioning** Complex scenes often exhibit heterogeneous motion patterns, where a single global deformation field fails to capture both large-scale rigid movements and fine non-rigid details. Motion-Aware Partitioning (MAPo) addresses this by recursively partitioning high-dynamic Gaussians into finer temporal segments, assigning dedicated deformation networks to each segment [31]. This hierarchical temporal partitioning allows for specialized motion modeling, capturing complex patterns better than a global field, albeit with increased storage costs. Similarly, DynaSplat employs a hierarchical motion modeling approach that blends multiple MLP modes for coarse global trends and fine local deformations, classifying Gaussians via temporal variance and optical flow consistency [8].

Relaxed Rigidity with Ray-based Grouping introduces a novel regularization strategy by clustering Gaussians intersected by identical view rays [32]. By applying motion coherence regularization within these groups, the method aligns displacement directions without constraining magnitudes, effectively isolating spatially contiguous surface Gaussians. This view-space grouping helps exclude occluded background primitives, improving reconstruction quality in dynamic regions.

**2.1.3 Handling Monocular and Sparse Inputs** Reconstructing dynamic scenes from monocular video introduces severe ambiguities regarding depth and scale. Methods like MotionGS integrate an optical flow decoupling module to separate camera ego-motion from object motion, using the resulting pure motion flow as a 2D prior to supervise Gaussian deformation [33]. This explicit guidance helps avoid noise from static backgrounds. For even more constrained settings, FreeGaussian derives dynamic Gaussian flow analytically from optical flow and camera motion, utilizing HDBSCAN clustering to isolate interactive objects without segmentation masks [34].

In sparse-view scenarios, SplatFields regularizes independent Gaussian primitives by predicting their attributes via implicit neural fields, enforcing local spatial bias through a tri-plane CNN generator [35]. This spatial autocorrelation stabilizes optimization when view coverage is insufficient. Neural Texture Splatting (NTS) further augments primitives with local RGBA tri-plane texture fields predicted by a shared global neural network, mitigating overfitting in sparse settings by enforcing spatial regularization across primitives [36].

## 2.2 4D Spatiotemporal Gaussian Primitives

An alternative to deformation fields is to explicitly model time as a fourth dimension within the Gaussian primitives themselves. These 4D Gaussian Splatting (4DGS) methods treat Gaussians as spatiotemporal entities, often parameterizing their evolution via basis functions or neural voxels.

**2.2.1 Neural Voxel and Grid Encodings** 4D-GS maintains a single set of canonical 3D Gaussians but employs a spatial-temporal structure encoder using decomposed 4D neural voxels (Hex-Plane) to extract features from center coordinates and timestamps [9]. A lightweight MLP decoder then predicts Gaussian deformations. This approach reduces storage complexity to  $O(N + F)$  by connecting adjacent time steps through the voxel grid. Similarly, DASH separates static and dynamic Gaussians, applying multiresolution 4D hash encoding exclusively to dynamic regions to bypass low-rank constraints and reduce hash table collisions [37]. Swift4D adopts a similar divide-and-conquer strategy, using a learnable parameter to decompose primitives and applying compact 4D hash encoding only to dynamic points [38].

To improve memory efficiency, MEGA decouples color into per-Gaussian DC components and shared AC MLP predictors, eliminating high-dimensional spherical harmonics [39]. It integrates

an entropy-constrained deformation field to predict position, scale, and rotation offsets, achieving significant storage reduction. 4D Scaffold Gaussian Splatting utilizes sparse 4D grid-aligned anchors containing learnable feature vectors, decoding them into sets of neural 4D Gaussians via shared MLPs [40]. This anchor-based compression maintains high primitive counts without storage penalties.

**2.2.2 Polynomial and Basis Function Parameterization** Instead of neural grids, some methods parameterize temporal evolution using analytical functions. Spacetime Gaussian Feature Splatting extends 3D Gaussians to 4D via temporal radial basis functions for opacity and polynomial trajectories for motion and rotation [41]. This captures emerging and vanishing content efficiently. Learnable Infinite Taylor Gaussian decomposes temporal evolution into a third-order Taylor polynomial for large-scale motion and a neural-network-predicted Peano remainder for high-frequency residuals [42]. This hybrid modeling balances the interpretability of polynomial trajectories with the flexibility of neural residuals.

Gaussian-Flow models dynamic scenes as deformable 3D Gaussian points using a Dual-Domain Deformation Model (DDDM), fitting time-dependent attribute residuals via joint polynomial and Fourier series functions [43]. This eliminates neural network overhead while preserving rendering speed. Similarly, Alias-free 4DGS derives a maximum sampling frequency formulation for 4D Gaussians based on the Nyquist-Shannon theorem, implementing a scale-adaptive filter to preserve anisotropy during deformation [44].

**2.2.3 Motion-Geometry Decoupling** A critical challenge in 4DGS is disentangling motion from geometry to prevent distortion. VeGaS applies a Galilean shearing matrix to the 4D covariance via congruence transformation, embedding time-varying velocity while preserving 3D conditional geometry through Schur complement invariance [10]. This allows independent trajectory modeling without distorting the underlying shape. Decoupling Motion and Geometry in 4DGS further emphasizes this by separating velocity anchors from geometric parameters, though this increases the parameter count [10].

## 2.3 Motion Factorization and Sparse Control

To reduce the complexity of modeling per-primitive motion, several methods factorize motion into shared bases or sparse control structures. This approach assumes that dynamic scenes often lie on a low-dimensional manifold of motion patterns.

**2.3.1 Trajectory Basis and Motion Fields** DynMF decomposes dynamic scene motion into a small set of globally shared learnable basis trajectories queried by a time-only MLP [11]. Each Gaussian stores motion coefficients blending these bases, significantly reducing optimization complexity. This sparsity-enforced coefficient blending enables explicit motion modeling with high efficiency. Similarly, 3DGS with Motion Trajectory Field models time-varying positions as linear combinations of DCT-initialized basis vectors weighted by MLP-predicted coefficients [45].

HAIF-GS initializes sparse motion anchors via farthest point sampling to drive Gaussian transformations through spatial interpolation [46]. An Anchor Filter predicts dynamic confidence scores to suppress redundant updates in static regions, reducing computational redundancy while maintaining global motion coherence. MotionScale represents dynamic scenes via canonical Gaussians governed by a scalable, cluster-centric motion field, partitioning dynamic Gaussians into disjoint clusters modeled by global rigid transformations and local non-rigid basis functions [47].

**2.3.2 Sparse Control Points and Anchors** Sparse-Controlled Gaussian Splatting (SC-GS) decomposes dynamic scenes into static canonical Gaussians and sparse control points [48]. A deformation MLP predicts time-varying 6-DoF transformations for these control points, and dense Gaussian motion fields are derived via interpolation. This reduces the parameter count significantly compared to per-Gaussian deformation. S4D (Streaming 4D) further decouples motion by generating 3D control points using ray coordinate systems, linking optical flow to 3D motion while learning hidden components via residual compensation [49].

Anchor-Driven Deformable and Compressed Gaussian Splatting (ADC-GS) organizes primitives into sparse anchors within a canonical space, employing a coarse-to-fine deformation strategy where anchors undergo global transformation via lightweight MLPs [50]. This grouping captures local rigidity and reduces deformation parameters. SD-GS replaces explicit Gaussians with a Canonical Gaussian Model comprising deformable anchor points, predicting spatiotemporal transformations via an Anchor Deformation Field [51].

## 2.4 Physics-Informed and Mesh-Hybrid Approaches

Incorporating physical laws or topological constraints can improve the plausibility and stability of dynamic reconstructions, especially in scenarios involving collisions, fluids, or deformable solids.

**2.4.1 Physics-Informed Learning** Neural Hamiltonian Deformation (NeHaD) replaces standard MLP deformation decoders with Hamiltonian Neural Networks (HNN) to learn conservative and solenoidal vector fields via scalar potential functions [12]. This ensures that Gaussian primitives evolve on symplectic manifolds, respecting energy conservation. Physics-Informed Deformable Gaussian Splatting (PIDG) models Gaussians as Lagrangian material points within a 4D decomposed hash space, enforcing the Cauchy momentum residual as a differentiable loss term to unify the modeling of fluids, solids, and rigid bodies [13].

FreeGave assigns a latent physics code to each Gaussian kernel, decoupling velocity estimation into linear and angular components using a divergence-free basis matrix parameterization [52]. This enables learning complex motion patterns without supervision. TRACE models each Gaussian kernel as a rigid particle, learning a translation-rotation dynamics system to estimate physical parameters like rotation center and velocity [53]. Spring-Gaus integrates a learnable 3D Spring-Mass system with Gaussian kernels, extracting anchor points to form a mass-spring network governed by stiffness and damping [54].

**2.4.2 Mesh-Gaussian Hybrids** Hybrid approaches combine the rendering flexibility of Gaussians with the topological constraints of meshes. MaGS constrains 3D Gaussians to roam near mesh surfaces via logical coordinates, enabling relative displacement during deformation [14]. This combines rendering flexibility with structural adaptability. Mesh-based Gaussian Splatting binds Gaussians to explicit mesh faces using barycentric coordinates and learnable normal offsets, enforcing topology-aware Gaussian splitting [15]. This provides necessary topological constraints for coherent deformation.

UniMGS unifies mesh and 3DGS with single-pass rasterization, treating depth-adjacent triangles as single entities for sub-pixel transmittance calculation [55]. It employs a Gaussian-centric binding strategy using ray casting to eliminate causal artifacts. BridgeSplat rigs 3D Gaussians to preoperative CT mesh triangles using optimizable barycentric coordinates, enforcing surface alignment

by fixing Gaussian means within parent faces [56]. This resolves monocular depth ambiguity in surgical navigation.

## 2.5 Domain-Specific Architectures

Different application domains impose unique constraints, leading to specialized architectural innovations.

**2.5.1 Autonomous Driving and Urban Scenes** Autonomous driving scenarios involve large-scale unbounded environments with rigid moving objects. DrivingGaussian decomposes scenes into static backgrounds and dynamic objects, constructing static backgrounds via Incremental Static 3D Gaussians and modeling moving objects via a Composite Dynamic Gaussian Graph [16]. AutoSplat enforces geometric flatness constraints on road and sky Gaussians to prevent floating artifacts during ego-vehicle shifts [17].

Street Gaussians assigns 3D Gaussians to each point in a neural point cloud, modeling dynamic vehicle appearance via 4D spherical harmonics using Fourier series [57]. HUGS decomposes scenes into static regions and rigidly moving dynamic objects represented by 3D Gaussians, parameterizing dynamic object poses via a unicycle model to regularize optimization under noisy tracking [58]. PAGS integrates offline semantic decomposition to assign binary criticality priors, employing a hybrid importance metric for pruning non-critical elements [59].

**2.5.2 Surgical and Medical Reconstruction** Surgical scenes involve deformable tissues and tool occlusions. SurgicalGS integrates multi-frame geometric data via binary motion masks to fuse depth-derived point clouds, employing a Flexible Deformation Model using Fourier and polynomial basis functions [18]. EndoGaussian utilizes Holistic Gaussian Initialization (HGI) to reproject pixels from predicted depth maps, bypassing slow SfM, and employs Spatio-temporal Gaussian Tracking (SGT) with decomposed HexPlane encoding [60].

ColorGS integrates Colored Gaussian Primitives utilizing dynamic anchors with learnable color parameters for spatially adaptive texture encoding [19]. SAGS (Self-Adaptive Alias-Free Gaussian Splatting) employs a self-adaptive weighted deformation decoder balancing attention and MLP branches to handle varying tissue deformation scales [61]. For colonoscopy, ColonSplat models colon motion via an MLP predicting time-dependent geometric offsets with clipping thresholds to enforce physical plausibility [62].

**2.5.3 Egocentric and Human-Centric Vision** Egocentric videos suffer from rapid camera motion and limited field of view. EgoGaussian partitions video into static and dynamic clips, initializing background and object shapes using masked 3DGS optimization with entropy regularization [63]. Photoreal Scene Reconstruction from an Egocentric Device integrates Visual-Inertial Bundle Adjustment (VIBA) to generate high-frequency continuous trajectories, correcting rolling-shutter timestamps [21].

For human avatars, Animatable 3D Gaussian models skinned Gaussians in canonical space, deforming them via linear blend skinning and employing a time-dependent ambient occlusion module for dynamic shadows [64]. GaussianBody decomposes dynamic modeling into canonical and observation spaces using SMPL-guided forward linear blend skinning [65]. HeadGaS extends 3DGS with per-Gaussian latent feature bases blended via expression weights for real-time animatable head avatars [66].

## 2.6 Robustness, Compression, and Streaming

Real-world deployment requires handling adverse conditions, reducing model size, and enabling online updates.

**2.6.1 Motion Blur and Adverse Conditions** Motion blur poses a significant challenge. BARD-GS decouples motion blur into camera and object components, optimizing virtual camera trajectories via  $SO(3)$  interpolation before modeling object motion through time-conditioned Gaussian deformation [26]. Unified Pose-aware Gaussian Splatting (UPGS) formulates camera and object motion as per-primitive  $SE(3)$  affine transformations, simulating motion blur by averaging rendered subframes [27]. Kinematics-GS constructs local kinematic priors to align Gaussian anisotropy with velocity trajectories, resolving monocular ambiguities in blurry scenes [67].

For low-light or thermal imaging, DarkGS learns a Neural Light Simulator (NeLiS) to calibrate camera-light systems, integrating calibrated illumination models into 3DGS [68]. Veta-GS integrates a view-dependent deformation field to model thermal variations, incorporating camera position and viewing direction [69]. UW-3DGS integrates a learnable underwater image formation module using voxel-based regression for spatially varying attenuation and backscatter [70].

**2.6.2 Compression and Efficiency** Model compression is critical for storage and transmission. SUCCESS-GS categorizes methods into Parameter Compression (pruning, quantization) and Restructuring Compression (anchor-based, neural integration) [22]. 4DGS-1K implements global pruning via a Spatial-Temporal Variation Score and key-frame based temporal filtering to eliminate short-lifespan Gaussians [23]. Temporally Compressed 3DGS (TC3DGS) integrates temporal relevance masking and gradient-aware mixed-precision quantization [71].

Lightweight 4D Gaussian Splatting (LGS) integrates Deformation-Aware Pruning and Gaussian-Attribute Pruning specifically for surgical scenes, achieving 15x compression [72]. Compact 3DGS implements learnable volume masking to prune redundant Gaussians during training and replaces spherical harmonics with hash-grid neural fields [73].

**2.6.3 Streaming and Online Reconstruction** Online reconstruction requires causal processing and temporal consistency. DASS (Dynamics-Aware Gaussian Splatting Streaming) partitions Gaussians into base and temporally-varied sets, employing a learnable selective inheritance mechanism to preserve relevant densified Gaussians [24]. Instant Gaussian Stream (IGS) employs an Anchor-driven Gaussian Motion Network to project multi-view 2D motion features into 3D space, decoupling motion prediction from appearance refinement [25].

HiCoM constructs a compact initial 3DGS via perturbation smoothing and employs a hierarchical coherent motion mechanism partitioning space into regions where Gaussians share translation and rotation parameters [74]. StreamSplat encodes frames into pixel-aligned static 3D Gaussians within an orthographic canonical space, predicting bidirectional deformation fields between consecutive frame embeddings [75].

## 3. Challenges and Outlook

Despite rapid progress, several critical challenges remain in dynamic Gaussian Splatting.

**Temporal Consistency and Long-Term Stability:** Many methods struggle with long-term temporal consistency, particularly in online or streaming settings. Error accumulation in defor-

mation fields or control point trajectories can lead to drift and artifacts over extended sequences [24], [74]. While sliding window approaches like SWinGS mitigate this by limiting canonical space scope [76], they introduce boundary inconsistencies. Future research must focus on global temporal regularization mechanisms that maintain consistency without sacrificing local adaptability.

**Physical Plausibility and Interaction:** While physics-informed methods like NeHaD [12] and PIDG [13] show promise, most dynamic GS methods rely on purely data-driven deformation fields that may produce physically implausible motions, especially under occlusion or in unobserved regions. Integrating robust physical simulators (e.g., rigid body dynamics, fluid dynamics) directly into the optimization loop remains an open problem. Furthermore, modeling complex interactions between multiple dynamic objects (e.g., hand-object interaction [77], [78]) requires advanced disentanglement strategies to prevent geometric interpenetration.

**Sparse and Monocular Ambiguities:** Reconstruction from monocular or sparse views remains ill-posed. Although methods leverage optical flow [33], depth priors [60], or foundation models [79], [80] to constrain the solution space, accuracy degrades significantly in textureless regions or under extreme motion. The reliance on external priors (e.g., SAM, optical flow networks) introduces latency and potential failure modes if these priors are inaccurate. Developing end-to-end methods that learn robust geometric priors from data without heavy reliance on external tools is a key direction.

**Scalability and Memory Efficiency:** Dynamic scenes, especially large-scale urban or indoor environments, require millions of Gaussians evolving over time. Even with compression techniques [22], [23], memory consumption remains a bottleneck for real-time applications on edge devices. Distributed training frameworks like DOGS [81] offer a path forward, but efficient inference and streaming of large dynamic scenes over networks require further innovation in level-of-detail (LOD) and progressive transmission strategies.

**Evaluation and Benchmarking:** The field lacks unified benchmarks for dynamic reconstruction. Metrics like PSNR and SSIM often fail to capture temporal flickering or physical plausibility. Recent efforts to evaluate egocentric reconstruction [20] and dynamic SLAM [82] are steps in the right direction, but comprehensive benchmarks covering diverse motion types, lighting conditions, and interaction scenarios are needed to fairly compare methodological families.

**Safety and Deployment:** In safety-critical applications like autonomous driving [16], [17] and surgical navigation [18], [19], the reliability of dynamic reconstruction is paramount. Artifacts such as floating geometry or incorrect motion prediction could lead to catastrophic failures. Ensuring robustness against sensor noise, adversarial inputs, and out-of-distribution scenarios is essential before widespread deployment.

## 4. Conclusion

Gaussian Splatting has revolutionized dynamic 3D scene reconstruction, offering a compelling balance of rendering speed, visual fidelity, and representational flexibility. From canonical deformation fields to 4D spatiotemporal primitives and physics-informed hybrids, the field has diversified rapidly to address the complexities of non-rigid motion, large-scale dynamics, and real-time constraints. While significant challenges remain in temporal consistency, physical plausibility, and scalability, the trajectory of research suggests a future where dynamic 3DGS enables immersive, interactive, and physically accurate digital twins of the real world. Continued integration of geometric priors, physical laws, and efficient compression strategies will be pivotal in advancing this



technology from laboratory benchmarks to real-world applications in robotics, entertainment, and scientific visualization.

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